



**Institut Mines-Télécom**

## **Solution of the exercises on high resolution methods**

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Contexte académique } **sans modifications**

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Séries chronologiques (partie 2) (TSIA202b)



# 1 Multiple Signal Classification (MUSIC)

**Question 1** It is sufficient to remark that

$$\mathbf{S}_k = \alpha_k \begin{bmatrix} 1 & z_k & \dots & z_k^{l-1} \\ z_k & z_k^2 & \dots & z_k^l \\ \vdots & \vdots & \ddots & \vdots \\ z_k^{n-1} & z_k^n & \dots & z_k^{N-1} \end{bmatrix} = \alpha_k \begin{bmatrix} 1 \\ z_k \\ \vdots \\ z_k^{n-1} \end{bmatrix} \begin{bmatrix} 1 & z_k & \dots & z_k^{l-1} \end{bmatrix}$$

**Question 2** Since  $s_t = \sum_{k=0}^{K-1} s_k[t]$ , we have  $\mathbf{S} = \sum_{k=0}^{K-1} \mathbf{S}_k$ , which leads to  $\mathbf{S} = \sum_{k=0}^{K-1} \alpha_k \mathbf{v}^n(z_k) \mathbf{v}^l(z_k)^\top$ . The factorization  $\mathbf{S} = \mathbf{V}^n \mathbf{A} \mathbf{V}^{l\top}$  is a simple rewriting of this last equality.

**Question 3** By using the equality  $\mathbf{S} = \mathbf{V}^n \mathbf{A} \mathbf{V}^{l\top}$ , we get the factorization  $\mathbf{R}_{ss} = \mathbf{V}^n \mathbf{P} \mathbf{V}^{nH}$ , where  $\mathbf{P} = \frac{1}{l} \mathbf{A} \mathbf{V}^{l\top} \mathbf{V}^{l*} \mathbf{A}^H$ . Thus matrices  $\mathbf{R}_{ss}$  and  $\mathbf{P}$  are Hermitian and positive semi-definite by construction. Moreover, matrix  $\mathbf{A}$  is invertible (because all  $\alpha_k$  are non-zero) and  $\mathbf{V}^l$  has full rank (because all poles are distinct), therefore  $\mathbf{P}$  is invertible, which proves that  $\mathbf{P}$  is positive definite. Moreover, since matrix  $\mathbf{V}^n$  has full rank (because all poles are distinct), we have  $\text{Span}(\mathbf{R}_{ss}) = \text{Span}(\mathbf{V}^n)$ , thus  $\text{rank}(\mathbf{R}_{ss}) = \text{rank}(\mathbf{V}^n) = K$ .

**Question 4** Since matrix  $\mathbf{R}_{ss}$  is Hermitian, it is diagonalizable in an orthonormal basis. Moreover, since it is positive semi-definite, its eigenvalues  $\{\lambda_i\}_{i=0 \dots n-1}$  are non-negative. Lastly, the fact that it has exactly rank  $K$  implies that

- $\forall i \in \{0 \dots K-1\}, \lambda_i > 0$ ;
- $\forall i \in \{K \dots n-1\}, \lambda_i = 0$ .

**Question 5**  $\mathbf{R}_{xx} = \mathbb{E} \left[ \frac{1}{l} \mathbf{X} \mathbf{X}^H \right] = \frac{1}{l} \mathbb{E} \left[ \mathbf{S} \mathbf{S}^H + \mathbf{S} \mathbf{B}^H + \mathbf{B} \mathbf{S}^H + \mathbf{B} \mathbf{B}^H \right]$ . However, since the noise is centered,  $\mathbb{E} \left[ \mathbf{S} \mathbf{B}^H \right] = \mathbf{S} \mathbb{E} \left[ \mathbf{B}^H \right] = \mathbf{0}$ . Similarly,  $\mathbb{E} \left[ \mathbf{B} \mathbf{S}^H \right] = \mathbf{0}$ , hence the result.

**Question 6**  $\mathbf{R}_{xx} \mathbf{w}_i = (\mathbf{R}_{ss} + \sigma^2 \mathbf{I}_n) \mathbf{w}_i = (\lambda_i + \sigma^2) \mathbf{w}_i$ , therefore  $\mathbf{w}_i$  is an eigenvector of  $\mathbf{R}_{xx}$  corresponding to the eigenvalue  $\lambda'_i = \lambda_i + \sigma^2$ .

**Question 7** For all  $i \in \{0 \dots K-1\}$ ,  $\mathbf{R}_{ss} \mathbf{w}_i = \lambda_i \mathbf{w}_i$ . Therefore  $\mathbf{w}_i = \frac{1}{\lambda_i} (\mathbf{V}^n \mathbf{P} \mathbf{V}^{nH}) \mathbf{w}_i$ . In particular,  $\mathbf{w}_i \in \text{Span}(\mathbf{V}^n)$ . Since this property holds for all  $i \in \{0 \dots K-1\}$ , we conclude that  $\text{Span}(\mathbf{W}) \subset \text{Span}(\mathbf{V}^n)$ . Moreover,  $\text{rank}(\mathbf{W}) = K$  because  $\mathbf{W}$  is an orthonormal matrix, hence linearly independent. In other respects, since  $\text{rank}(\mathbf{V}^n) \leq K$ , we conclude that  $\text{Span}(\mathbf{W}) = \text{Span}(\mathbf{V}^n)$ .

**Question 8** For all  $k \in \{0 \dots K-1\}$ ,  $\mathbf{v}^n(z_k) \in \text{Span}(\mathbf{V}^n)$ . However,  $\text{Span}(\mathbf{V}^n) = \text{Span}(\mathbf{W}) \perp \text{Span}(\mathbf{W}_\perp)$ . We conclude that  $\mathbf{v}^n(z_k) \perp \text{Span}(\mathbf{W}_\perp)$ , therefore  $\|\mathbf{W}_\perp^H \mathbf{v}^n(z_k)\|^2 = 0$ . Thus, all poles  $\{z_k\}_{k \in \{0 \dots K-1\}}$  are solutions. If there were another solution  $z$  distinct from the  $z_k$ , we would then get  $\mathbf{v}^n(z) \perp \text{Span}(\mathbf{W}_\perp)$ , therefore  $\mathbf{v}^n(z) \in \text{Span}(\mathbf{V}^n)$ . This is impossible because the set of vectors consisting of the columns of  $\mathbf{V}^n$  and of  $\mathbf{v}^n(z)$  would be linearly independent if  $z$  were different from the  $z_k$ . In conclusion, the poles  $\{z_k\}_{k \in \{0 \dots K-1\}}$  are the unique solutions of the equation  $\|\mathbf{W}_\perp^H \mathbf{v}^n(z)\|^2 = 0$ .



## 2 Estimation of Signal Parameters via Rotational Invariance Techniques (ESPRIT)

**Question 1** We get  $D = \text{diag}(z_0, z_1, \dots, z_{K-1})$ .

**Question 2** Since matrices  $V^n$  and  $W$  are two bases of the same subspace, they satisfy the relation  $V^n = W G$ , where  $G$  is the transfer matrix from basis  $W$  to basis  $V^n$ . The equalities  $V_{\downarrow}^n = W_{\downarrow} G$  and  $V_{\uparrow}^n = W_{\uparrow} G$  are obtained by respectively extracting the  $n - 1$  first rows and the  $n - 1$  last rows of the equality  $V^n = W G$ .

**Question 3**  $W_{\uparrow} = V_{\uparrow}^n G^{-1} = V_{\downarrow}^n D G^{-1} = W_{\downarrow} G D G^{-1}$ . We thus get the equality  $W_{\uparrow} = W_{\downarrow} \Phi$ , where  $\Phi = G D G^{-1}$ . This is the diagonalized form of matrix  $\Phi$ . Thus the eigenvalues of  $\Phi$  are the diagonal entries of  $D$ , i.e. the poles  $z_k$ .

**Question 4** Since  $W_{\uparrow} = W_{\downarrow} \Phi$ , we have  $W_{\downarrow}^H W_{\uparrow} = W_{\downarrow}^H W_{\downarrow} \Phi$ , thus  $\Phi = (W_{\downarrow}^H W_{\downarrow})^{-1} W_{\downarrow}^H W_{\uparrow}$ .

**Question 5** The ESPRIT method is composed of several steps :

- compute the estimator  $\hat{R}_{xx}$  of matrix  $R_{xx}$ ,
- diagonalize it to obtain an estimation of matrix  $W$ ,
- extract from  $W$  the matrices  $W_{\downarrow}$  and  $W_{\uparrow}$ ,
- compute  $\Phi = (W_{\downarrow}^H W_{\downarrow})^{-1} W_{\downarrow}^H W_{\uparrow}$ ,
- diagonalize matrix  $\Phi$  to obtain the poles  $\{z_k\}_{k \in \{0 \dots K-1\}}$ .



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